VRANDESH BANDIKATTI

SENIOR DATA AND PLATFORM ENGINEER | MICROSOFT FABRIC | AZURE | DATABRICKS | PYTHON/SQL | DATA ENGINEERING
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PROFILE

Senior Full Stack Tech Lead and Engineering leader with 16+ years of software engineering experience designing and delivering production SaaS platforms across enterprise data, payments, workflow automation and cloud-native applications. Strong hands-on background in Python, FastAPI, Django, REST/GraphQL APIs, Fabric Integration, Embedded PowerBI and Fabric Integration, distributed services, event-driven architectures and workflow orchestration.

I have spent the last several years designing governed enterprise data and processing platforms across Azure, Fabric, Databricks, Python and Power BI, while also bringing unusually strong software engineering, DevOps and AI-readiness capability.

Experienced building production systems incorporating asynchronous processing, stateful workflows, retries, idempotency, rate limiting, caching, authentication, observability and automated CI/CD. Delivered microservice and serverless architectures across GCP, AWS and Azure, including Cloud Functions/Firebase, AWS Lambda/Step Functions and Azure Durable Functions.

Built and maintained 12+ production microservices and complex enterprise integrations across financial, data and analytics platforms. Recent work includes Python-based orchestration pipelines, FastAPI services, high-volume document-processing workflows, scheduled data processing, governed data integrations and automated production workloads.

Strong understanding of cloud-native engineering, Docker-based deployment, API integration, data pipelines and backend modernization, with experience working across SQL/NoSQL data stores, enterprise data platforms and event-driven systems. Comfortable owning backend architecture while remaining hands-on with implementation, testing, deployment, troubleshooting and production support.

DATA ENGINEERING & CLOUD PLATFORMS

Data & Integration Engineering — Enterprise integrations across SQL/NoSQL databases, data platforms and third-party APIs including Fabric One Lake, Power BI, Databricks, Microsoft Fabric, Xero, MYOB and Workday Adaptive Planning. Batch, scheduled and event-driven processing patterns.

Security — OAuth/OIDC-aligned identity patterns, Entra ID, RBAC, service principals, secrets management, access-controlled API/data processing and least-privilege system design.

Python & Backend Engineering — Python 3.x, FastAPI, Django, asynchronous/backend processing, Node.js, Next.js, REST APIs, GraphQL, service-layer design, validation, structured data processing, caching, rate limiting and production error handling.

Microservices & Distributed Systems — Designed and maintained 12+ production microservices and integration services. Event-driven and serverless patterns, service boundaries, API contracts, retries, idempotency, graceful degradation and third-party system integration.

Cloud Engineering — GCP: Cloud Functions, Firebase, Firestore and serverless workloads. AWS: Lambda, Step Functions and event-driven architectures. Azure: Functions, Durable Functions, Azure SQL, Blob Storage, Databricks, Data Factory and Key Vault. Docker and automated DEV/UAT/PROD deployment.

DevOps & Reliability — Docker based deployment, Git-based delivery, CI/CD, environment promotion, automated deployment, production observability, Grafana, tracing, alerting, logging and incident support.

GEN AI/Agentic Systems — Multi-agent orchestration, LangGraph state machines, LangChain, tool/function calling, MCP tool integration, structured output contracts (Pydantic / JSON schema), prompt engineering, self-repair and bounded retry loops, human-in-the-loop approval gates, Azure OpenAI, Amazon Bedrock, Claude API

Rag & Knowledge engineering — Hybrid retrieval (vector + BM25), semantic chunking, embeddings and vector stores, re-ranking, context assembly with source citations, grounded generation, hallucination detection, PII and policy guardrails, evaluation harnesses and human feedback loops, Grafana and LangSmith observability.

Workflow Orchestration — Production workflow design using Azure Durable Functions and AWS Step Functions, including parallel execution, state management, retries, idempotent activities, scheduled workloads, failure recovery and long-running workflow coordination.

EDUCATION, CERTIFICATIONS & IP

- Harvard Business School Online — Certificate in Business Analytics, 2023
- BE, Electronics & Communication — Visvesvaraya Technological University, First Class Honours, 2008
- Certifications — Azure for Data Engineering (Coursera); Machine Learning & Big Data; Generative AI; React with TypeScript. PMI Melbourne member. PMP, Anthropic Claude Developer, Snowflake and Databricks pathways in progress.

PROFESSIONAL EXPERIENCE

Senior Full Stack Developer - SaaS Platform Architecture — Findex Australia

Melbourne · March 2022 – Present

- Architected and delivered production backend services and workflow-driven enterprise SaaS platforms using Python, Django, FastAPI, GraphQL, REST APIs and cloud-native services.
- Designed distributed processing workflows using Azure Durable Functions with parallel execution, retry policies, state tracking and idempotent activities for high-volume regulatory document processing.
- Built Django and FastAPI backend services with authentication, authorization, caching, rate limiting, structured query execution, bounded failure recovery, tracing and production alerting.
- Designed Python processing pipelines for complex financial, payroll and regulatory workloads, combining structured extraction, deterministic business rules, validation and exception processing.
- Established API and integration patterns across Power BI, Databricks, Microsoft Fabric, SharePoint, Workday Adaptive Planning, Xero, MYOB and internal enterprise services.
- Designed scheduled and automated data-processing workflows for financial modelling, benchmarking, student analytics and board-report generation.
- Defined engineering standards across API integration, service boundaries, CI/CD, security, observability, environment promotion and production support.
- Led cross-functional engineering teams while remaining hands-on with Python, Django, FastAPI, GraphQL, cloud integration and backend architecture.
- Stepped in as interim DevOps lead during a critical production incident, stabilising backend services, improving CI/CD reliability, adding Grafana observability and reducing manual operational overhead.

Senior Software Engineer, Product Innovation, Backend Cloud Engineering— Quest Payment Systems

Melbourne · October 2017 – March 2022

- Led backend and cloud engineering from concept through production for Airpay App, a regulated POS App and Donation Point Go, a regulated contactless donation SaaS platform.
- Architected and maintained 12+ production microservices and integration services using Node.js, AWS Lambda and AWS Step Functions.
- Designed event-driven, auto-scaling serverless architectures across AWS and GCP to handle unpredictable payment and transaction workloads.
- Built third-party integrations with Xero, MYOB and payment ecosystem services using resilient API integration patterns.
- Developed backend APIs supporting web, mobile, POS and kiosk applications.
- Implemented Google Firebase Functions, Firestore/real-time data services and NoSQL persistence for Airpay and Donation Point platforms.
- Led a team of three while remaining hands-on with backend development, cloud architecture, monitoring, logging and production support.

Senior Full Stack Developer – ESRI Australia

Melbourne · August 2017 – September 2017

- Developed enterprise geospatial and ArcGIS-based applications supporting location-based business solutions.
- Built AngularJS front-end modules for interactive enterprise applications.
- Integrated geospatial and enterprise services using SOAP/WSDL APIs.
- Developed Node.js / Express middleware to support application and service integration.
- Supported AWS-hosted infrastructure using EC2, Elastic Beanstalk, SQS and SNS.
- Implemented asynchronous processing and messaging workflows for distributed application components.

Contributed across full-stack development, cloud deployment, systems integration and production support.

Technology Lead / Full Stack Developer — Apptarix Mobility

Bangalore · June 2013 – July 2017

- Filed a provisional patent application 201641021557 — a recommendation engine combining commercial and learnt rules, with a method to measure which of the two is earning its place.
- Built an early NLP sentiment engine using cosine similarity in R and Python; integrated clustering and sentiment analytics into OTT, analytics and EdTech product lines. Managed a team of 5. Integrated R and Python analytics for clustering and sentiment analysis, converting raw user data into product intelligence and customer insight.
- Designed scalable data ingestion, scraping and processing services that expanded the product data ecosystem and supported analytics-driven features.
- Managed a team of 5 in a startup environment using Agile and Scrum; led architecture across analytics dashboards, OTT SaaS platforms and client-facing product lines including Teletango, PrepSuite and MystiKit.

BACKEND, WORKFLOW & DATA PLATFORMS DELIVERED

KRM Remuneration Platform — SaaS Platform - Secure Data Processing & Calculation Services

Findex · Platform Architect and Delivery Lead

Architected the backend, data and security foundations for an enterprise remuneration platform processing sensitive financial, incentive and workforce data.

- Designed processing pipelines for short-term incentive, referral, forecasting and audit workloads across enterprise financial datasets.
- Implemented Entra ID SSO, RBAC and row-level security across application services and Power BI Embedded.
- Replaced spreadsheet-based remuneration calculations with a deterministic, auditable processing engine.
- Designed backend authorization controls that determine permitted data access before sensitive information enters downstream processing workflows.
- Established governed bronze, silver and gold data layers using Databricks, Microsoft Fabric and OneLake.
- Built the underlying remuneration calculation engine covering performance weights, thresholds, ramps, payaway curves, gateways, caps, floors and employee segmentation. Integrated application, analytics and data-platform components across controlled DEV, UAT and production environments.
- Supported production onboarding of 443 users and 129 people leaders under the platform's role and permission model.

Payroll AI — Python Workflow Automation & Payroll Assurance SaaS Platform

Findex · Solution Architect and Hands-on Engineer

Designed and built a Python-based multi-stage processing platform that converts Modern Award and Enterprise Agreement documents into executable payroll rules, validates those rules against payroll data and surfaces exceptions through Power BI.

- Designed a four-stage workflow covering document ingestion, structured rule extraction, client-specific calculation generation, payroll execution and exception remediation.
- Built the processing pipeline in custom Python, including retrieval and chunking strategies designed to preserve legal clause references across dense regulatory documents.
- Implemented traceability from every calculated payroll outcome back to its originating legal clause, supporting explainability, auditability and business validation.
- Designed controlled change workflows that determine the blast radius of a rule update across pay runs and clients before material changes are approved.
- Implemented automated exception classification and rerun coordination, reducing the need for developers to manually intervene in individual payroll corrections.
- Converted under- and over-payment detection from periodic audit activity into a continuous processing and exception-management workflow with human approval for flagged cases.

WealthAI — Distributed Document Processing & Workflow Orchestration

Findex · Solution Designer and Full Stack Development

Designed a serverless processing platform for concurrently processing large volumes of regulatory documents using stateful orchestration and independently executable workflow activities.

- Designed end-to-end workflow orchestration using Azure Durable Functions, including parallel execution, dependency management, retries and persistent workflow state.
- Separated probabilistic AI extraction from deterministic business-rule execution so regulatory outcomes remained repeatable and explainable. Integrated workflow outputs with JSON, Excel and database services. Designed graceful-degradation and failure-handling patterns so individual document failures did not compromise the complete processing workload.

PerformX — FastAPI Backend & Governed Data Query SaaS Platform

Findex · Solution Architect and Hands-on Backend Engineer

Designed and delivered a production FastAPI service that converts natural-language requests into validated queries against governed enterprise data, enforcing authentication, authorization and data-access policies before execution.

- Built backend APIs using FastAPI to handle identity resolution, query generation, validation, execution and response orchestration.
- Implemented Entra ID authentication and permission enforcement across multiple stages of request processing to prevent unauthorized access to financial data.
- Designed bounded retry and self-recovery logic where failed generated queries are corrected using runtime errors, revalidated and terminated safely when recovery is unsuccessful.
- Ensured authoritative values are always produced by the governed Power BI semantic model rather than generated by the language model.